DESIGN MILESTONES

EGG DROP CHALLENGE

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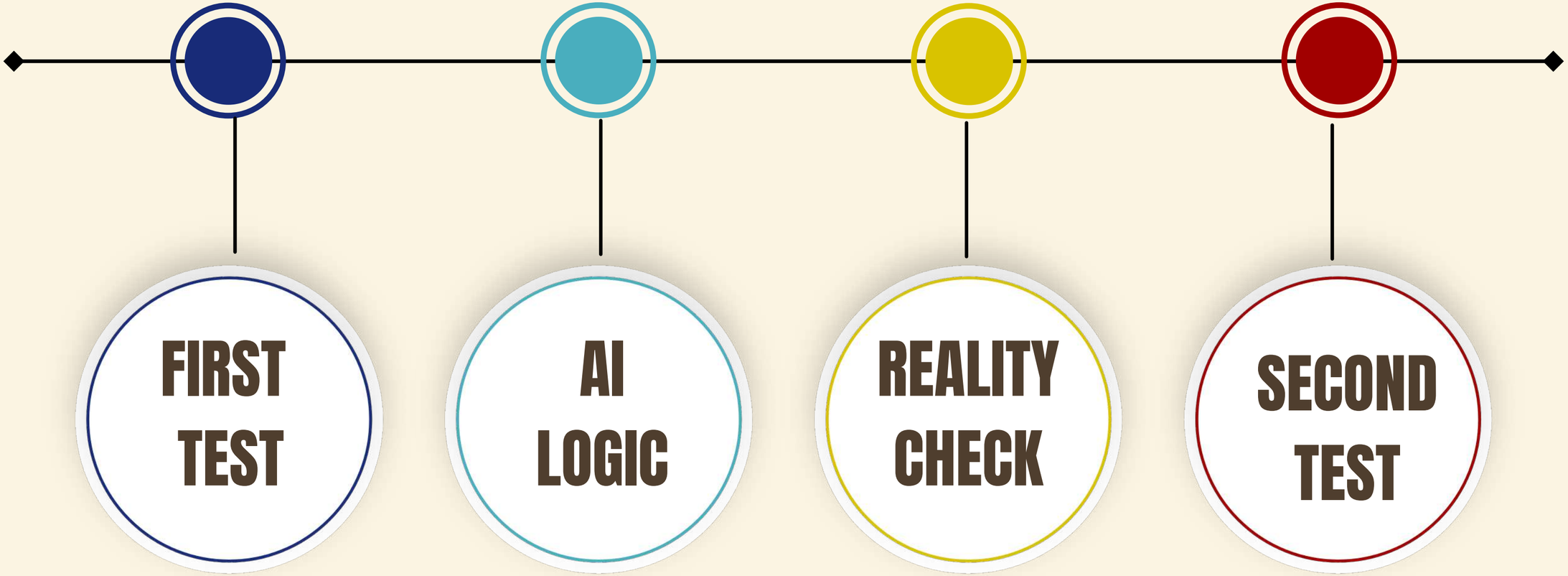
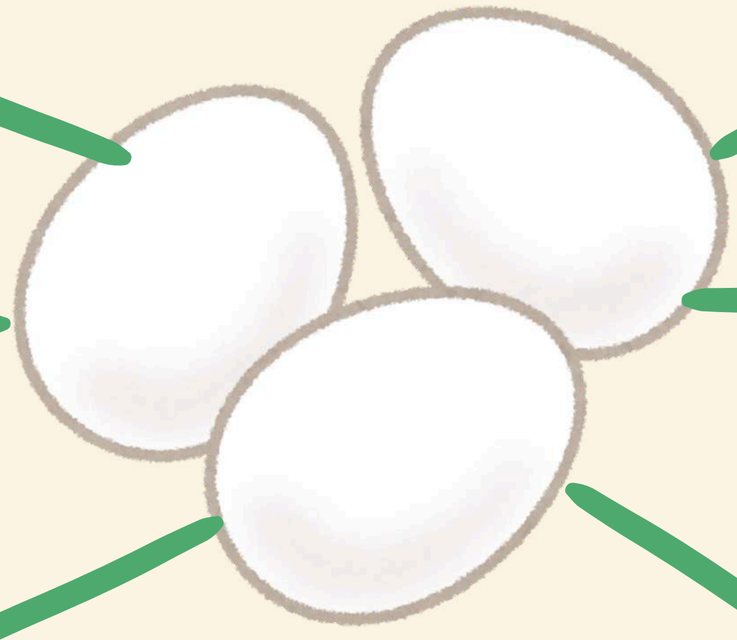
PROCESS VISUALIZATION



KEYWORDS



SUSPENSION

TENSEGRITY

CRUMPLE ZONE

TENSION

COMPRESSION

CENTER OF GRAVITY

AI ROLE TAGS

ERROR FINDER

- Weak Spots: warned me about the straw joints
- Bending Alert: Predicted where the straws would snap

IDEA CREATOR

- New Strategy: changed from hanging the egg to wrapping
- String Balance: used tension to soak up the shock

SHAPE DESIGNER

- Balance: kept the structure steady during the fall
- Space Saver: fit the whole design in the 15cm limit

THE LOGIC HELPER

- Explained why my first design failed
- Explained g-force
- Taught me how to impact moves through flaws

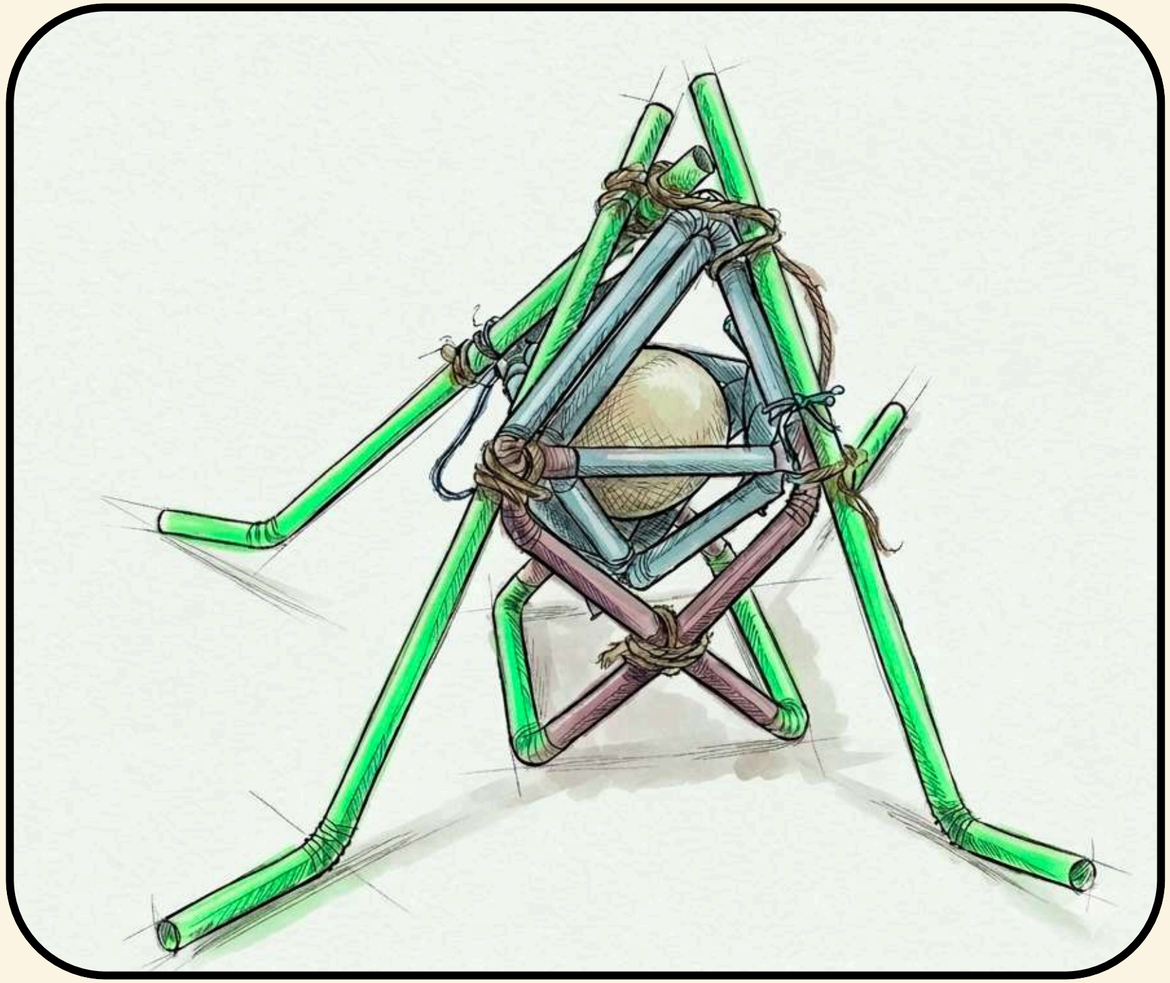
PHASE 1

WHAT I ACCEPTED

- SUSPENSION SYSTEM
- TRIANGULAR FRAME
- TENSION BALANCE
- TOP HEAVY SET UP

WHAT I REJECTED

- GLUED JOINTS
- FULL TAPE WRAP
- COILED STRAWS
- DOUBLE FRAME



REFLECTION

In Phase 1, I used my own logic and did not use AI at all. I designed a simple **hexagonal frame** because I believed hanging the egg in the center was the best way to keep it safe. While the design **survived a 1.5-meter drop**, it had a high center of gravity and weak legs, which made it unstable.

Because of this lack of balance, the structure couldn't handle the pressure and t**he egg broke at 3 meters**. This phase was a lesson in my own human intuition and the limits of manual construction.

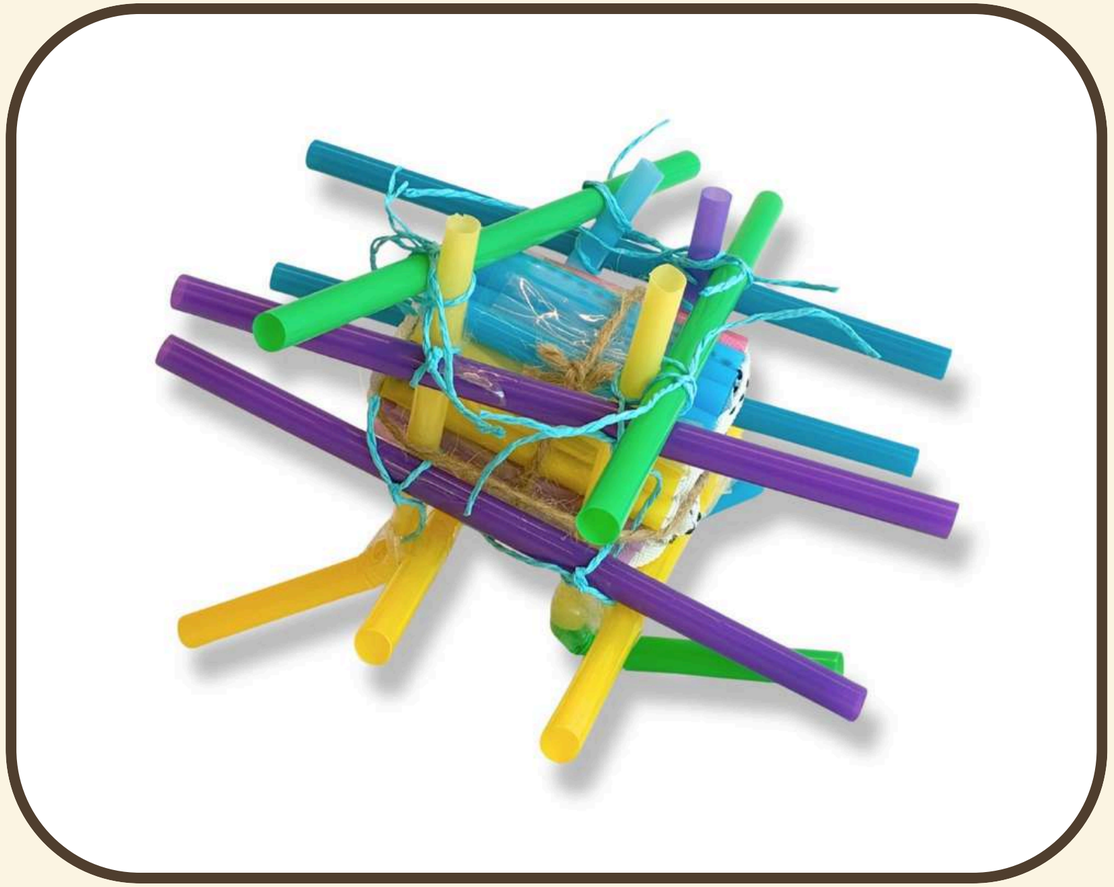
PHASE 2

WHAT I ACCEPTED

- DOUBLE BASE
- TIGHTER STRAWS
- LOW BALANCE

WHAT I REJECTED

- VERTİCAL LEGS
- ELASTIC STRING
- DOUBLE EGG CRADLE



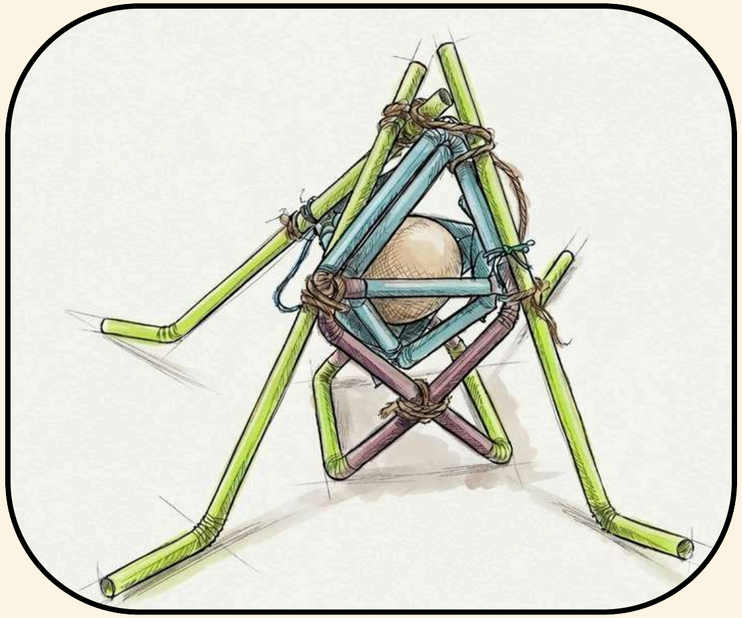
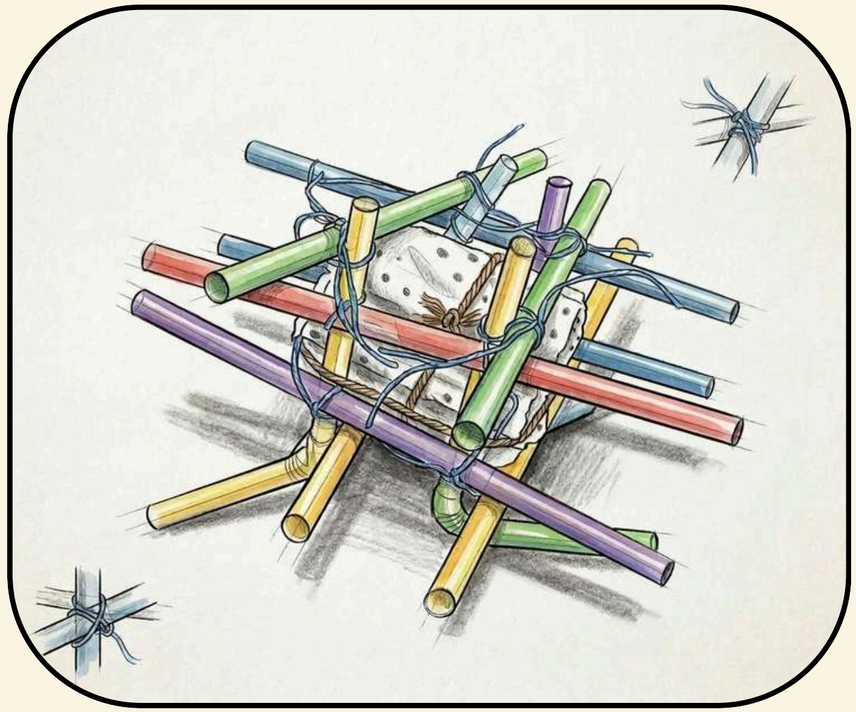
REFLECTION

After the egg broke from 3 meters in Phase 1, I used AI to solve the **physics issues of balance and strength.** The AI helped me **lower the center of gravity** and **reinforce the weak legs** of the frame to handle the higher impact. By applying these physical logic changes, I created a much more protected and stable structure. This time, the design was perfectly balanced, and the **egg survived the 3-meter drop without a single crack.**

WHAT I LEARNED

- **Failure is the Best Teacher:** Phase 1 taught me that human intuition alone can miss small physics details like the center of gravity. A high center of gravity and weak legs led to a 3-meter failure.

- **AI as a Problem Solver:** In Phase 2, I learned that AI is most powerful when used to debug a problem. It helped me understand exactly why the legs buckled and how to reinforce them.

- **Human-AI Synergy:** The ultimate lesson was that a successful design needs both my original creative idea and the AI's analytical stress-testing.

- **Balance is Everything:** I realized that "protection" isn't just about padding; it's about stability and balance during the flight and the final impact.